

Exponential functions



1 Would an exponential function generate the values shown in the following tables?

a

x	1	2	3	4	5	6
$f(x)$	5	25	125	625	3125	15 625

b

x	1	2	3	4	5	6
$f(x)$	4	9	11.5	15	13.5	11

2 What type of function would generate values as shown in the table?

x	1	2	3	4	5	6
$f(x)$	2	3	4.8	7.6	11.8	18

A Approximately exponential **B** Exactly exponential **C** Not exponential

3 Determine whether the following are increasing or decreasing exponential functions:

- a** $y = 10 \times \left(\frac{3}{5}\right)^x$ **b** $y = 9 \times 3^x$ **c** $y = 3 \times (0.5)^x$ **d** $y = 0.2 \times 2^x$
e $y = 2.2 \times 1.05^x$ **f** $y = 50 \times \left(\frac{7}{4}\right)^x$ **g** $y = 500 \times 0.75^x$ **h** $y = 80 \times 2^{-x}$
i $y = -5 \times 2^x$ **j** $y = -0.5 \times 0.8^x$ **k** $y = -2 \times \left(\frac{1}{4}\right)^x$ **l** $y = -4 \times 3^{-x}$

4 Which one of the following exponential functions rises most steeply?

A $y = 2 \times (2.1)^x$ **B** $y = 2 \times (2.2)^x$ **C** $y = 2 \times (1.7)^x$ **D** $y = 2 \times (1.2)^x$

5 For the following pairs of functions, select the function that increases more rapidly for $x > 0$:

a $y = 4^x$ and $y = 5^x$ **b** $y = 2^x$ and $y = 3 \times 2^x$

6 Find the missing coordinate in each ordered pair so that the pair is a solution of $y = 3^{-x}$:

a $(0, \text{?})$ **b** $(\text{?}, 9)$ **c** $(-1, \text{?})$ **d** $\left(\text{?}, \frac{1}{27}\right)$



7 Find the missing coordinate in each ordered pair so that the pair is a solution of $y = -3^x$:

a $(5, 1)$

b $\left(1, -\frac{1}{27}\right)$

c $(-1, 1)$

d $(1, -81)$

8 Consider the function $f(x) = 3 + 4^{x+5}$. Evaluate $f(-2)$.

9 Consider the function $f(x) = \left(\frac{4}{3}\right)^x$.

a Evaluate $f(0)$.

b Evaluate $f(-2)$.

10 Find the value of $5000(1.08^x)$ for $x = 30$ to two decimal places.

11 Find the value of $200\left(\frac{1.03^n - 1}{0.03}\right)$ for $n = 24$ to two decimal places.

12 Find the value of $P\left(1 + \frac{r}{k}\right)^{kn}$ for $P = 3000$, $r = 5\%$, $k = 4$ and $n = 2$ to two decimal places.

13 Consider the expression 2^x .

a Evaluate the expression when:

i $x = -4$

ii $x = 0$

iii $x = 4$

b For what values of x will 2^x have a negative value?

c Describe what happens to the value of 2^x as x gets larger.

d Describe what happens to the value of 2^x as x gets smaller.

14 Consider the function $y = 3^x$.

a Complete the table of values:

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y												

b Describe the behavior of the function as x increases.

c State the domain of the function.

d State the range of the function.

15 Consider the function $y = 2^{-x}$.



a Complete the table of values:

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y												

b Describe the behavior of the function as x increases.

c Find the y -value of the y -intercept of the function.

d State the domain of the function.

e State the range of the function.

16 Consider the function $y = -(2^x)$.

a Complete the table of values:

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
y											

b Can the value of y ever be zero or positive? Explain your answer.

c Is $y = -(2^x)$ an increasing or decreasing function?

d Describe the behavior of the function as x increases.

e State the domain of the function.

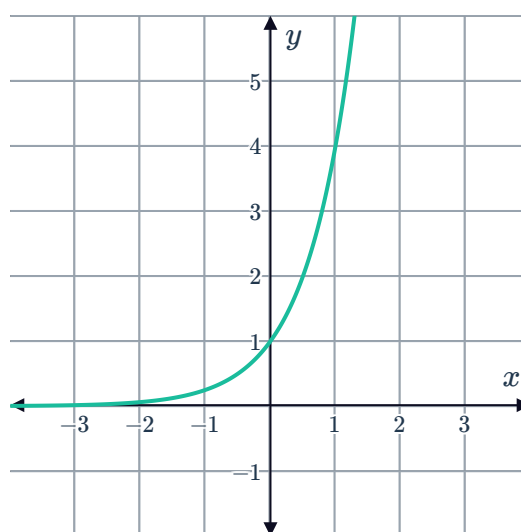
f State the range of the function.

17 Consider the graph of the equation $y = 4^x$:

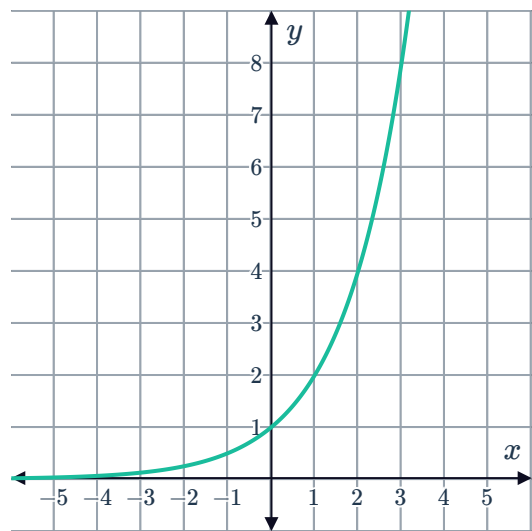
a Is each y -value of the function positive or negative?

b State the value of y the graph approaches but does not reach.

c State the equation and name of the horizontal line, which $y = 4^x$ gets closer and closer to but never intersects.



18 The graph of $y = 2^x$ is shown:



- a State the y -intercept of this graph.
- b Does the graph have an x -intercept?
- c State the domain of the function.
- d State the range of the function.
- e Find the value of y when $x = 7$.
- f Find the value of x when $y = 256$.

19 Consider the function $f(x) = e^x$.

- a Complete the following table of values, correct to two decimal places:

x	-3	-2	-1	0	1	2	3
$f(x)$							

- b Sketch the graph of $f(x) = e^x$.

20 Consider the function $y = -10^x$.

- a Complete the following table:

x	-2	-1	0	1	2	3
y						

- b For any value of x , is y always negative for this equation? Explain your answer.
- c Sketch the graph of the function.

21 Consider the function $y = 8^{-x}$.

- a Can the value of y ever be zero or negative? Explain your answer.
- b As the value of x gets larger and larger, what value does y approach?
- c As the value of x gets smaller and smaller, what value does y approach?
- d Find the y -value of the y -intercept.
- e How many x -intercepts does the curve have?
- f Sketch the graph of $y = 8^{-x}$.

22 Consider the function $y = 4(2^x)$.



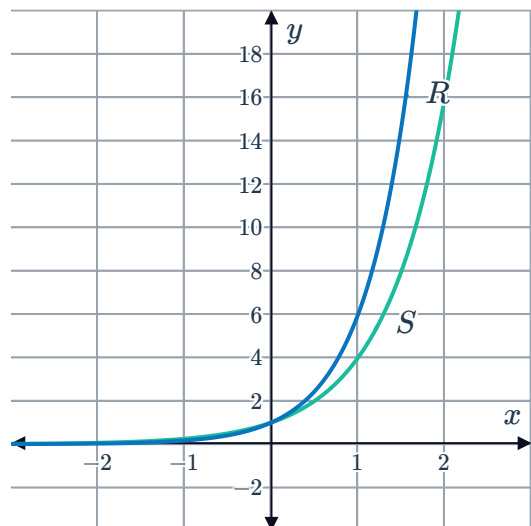
- a** Find the y -value of the y -intercept.
- b** Can the function value ever be negative?
- c** As x approaches infinity, what value does y approach?
- d** Sketch the graph of $y = 4(2^x)$.

23 Consider the function $y = 2(3^x)$.

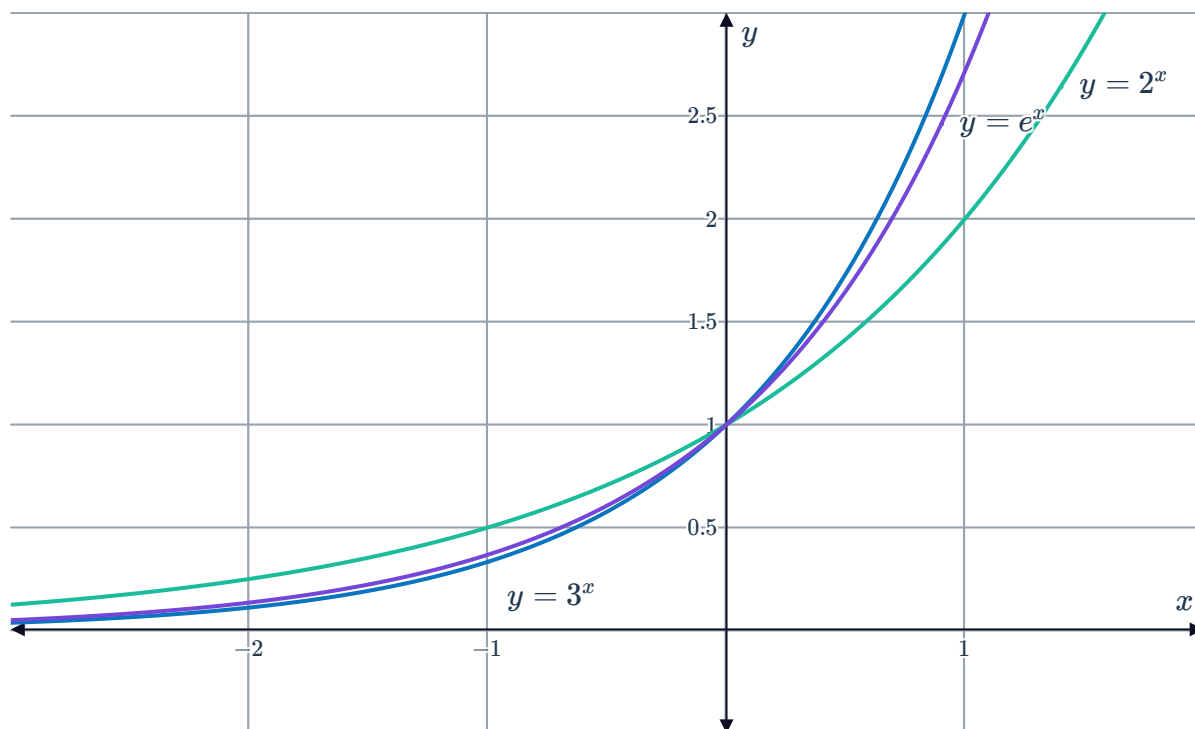
- a** Find the y -value of the y -intercept.
- b** Can the function value ever be negative? Explain your answer.
- c** State the domain.
- d** State the range.
- e** As x approaches positive infinity, what value does y approach?
- f** Sketch the graph of $y = 2(3^x)$.

24 Consider the graphs of the two exponential functions R and S :

- a** One of the graphs is of $y = 4^x$ and the other graph is of $y = 6^x$. Which is the graph of $y = 6^x$?
- b** For $x < 0$, is the graph of $y = 6^x$ above or below the graph of $y = 4^x$. Explain your answer.



25 The functions $y = 2^x$, $y = e^x$ and $y = 3^x$ have been sketched on the same coordinate plane:

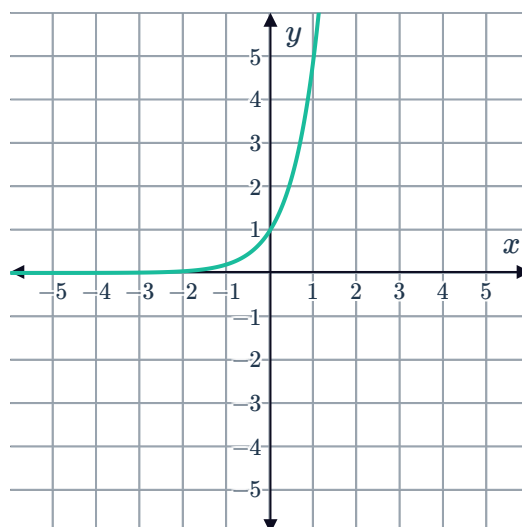


- a For what values of x does the inequality $3^x > e^x$ hold?
- b For what values of x does the inequality $2^x > e^x$ hold?

Transformations of exponential functions

26 Consider the given graph of $y = 5^x$.

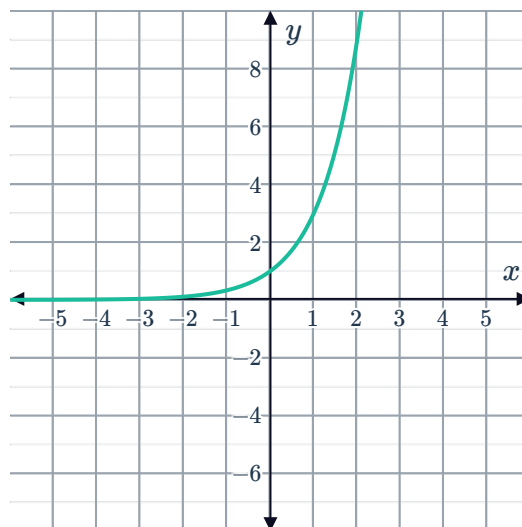
- a Describe a transformation of the graph of $y = 5^x$ that would obtain $y = -5^x$.
- b Sketch the graph of $y = -5^x$.



27 Consider a graph of $y = 3^x$:

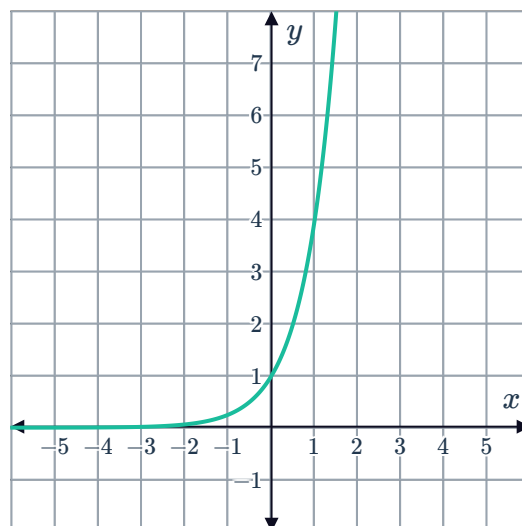


- a Describe a transformation of the graph of $y = 3^x$ that would obtain the graph of $y = 3^x - 4$.
- b Sketch the graph of $y = 3^x - 4$.



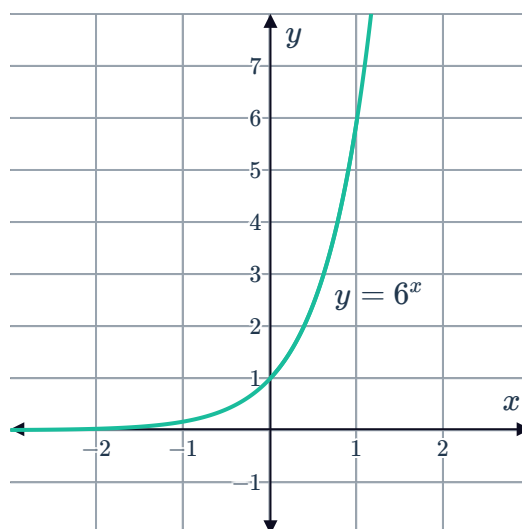
28 Consider the given graph of $y = 4^x$:

- a Describe a transformation of the graph of $y = 4^x$ that would obtain $y = 4^{-x}$.
- b Sketch the graph of $y = 4^{-x}$.

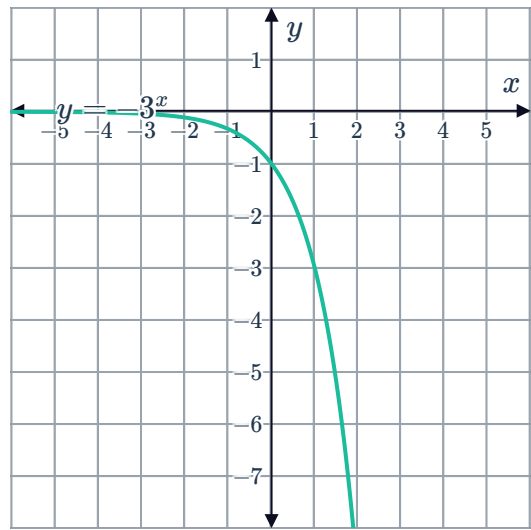


29 Consider the given graph of $y = 6^x$:

- a Describe a transformation of the graph of $y = 6^x$ that would obtain $y = -6^{-x}$.
- b Sketch the graph of $y = -6^{-x}$.



30 Consider the given graph of $y = -3^x$:



- a State the asymptote of $y = -3^x$.
- b Find the asymptote of $y = 2 - 3^x$.
- c How many x -intercepts would $y = 2 - 3^x$ have?
- d State the domain of $y = 2 - 3^x$.
- e State the range of $y = 2 - 3^x$.

31 Consider the functions $y = 2^x$ and $y = 2^x - 2$.

- a Find the y -intercept of $y = 2^x$.
- b Determine the y -intercept of $y = 2^x - 2$.
- c State the horizontal asymptote of $y = 2^x$.
- d Determine the horizontal asymptote of $y = 2^x - 2$.

32 For each of the following functions, determine the following:

- i Horizontal translation
- ii Vertical translation
- iii Growth constant

a $y = 5^{x+4} + 2$ b $y = 5^{3x+3} - 1$ c $y = 2^{5x-1} + 2$

33 Find the equation of the new function that results from the following transformations:

- a The graph of $y = 2^x$ is translated down by 7 units.
- b The graph of $y = 11^x$ is moved to the left by 10 units.

34 For each of the following, write the equation of the new function that results from $y = 7^x$ after the following translations:

- a Translated three units to the right and then five units downward.
- b A reflection in the x -axis, then translated five units upward and three units to the left.
- c Translated five units upward, three units to the left and then reflected in the x -axis.

35 Sketch the graph of the following functions:

a $y = 2^{x+4}$ b $y = 2^x + 4$



36 For each of the following exponential functions:

- i Find the y -value of the y -intercept.
- ii State the domain.
- iii State the range.
- iv Sketch the graph of the function.

a $y = 4^x + 3$

b $y = 2^x - 5$

c $y = 2^{x+4}$

d $y = 3^{x-1}$

e $y = 3 - 2^{-x}$

37 Consider the function $y = -5^x$.

- a** State the equation of the asymptote of $y = -5^x$.
- b** State the equation of the asymptote of $y = 2 - 5^x$.
- c** How many x -intercepts would the graph of $y = 2 - 5^x$ have?

38 For each of the following functions:

- i Find the y -intercept of the curve.
- ii State the equation of the horizontal asymptote.
- iii Sketch the graph of the function.

a $y = 3^x + 2$

b $y = 2^x - 2$

c $y = -3^x + 2$

d $y = 3^{-x} - 1$

e $y = 3^{-x} + 1$

39 Consider the original graph $y = 3^x$. The function values of the graph are multiplied by 2 to form a new graph.

- a** For each point on the original graph, find the point on the new graph:

Point on original graph	Point on new graph
$\left(-1, \frac{1}{3}\right)$	$(-1, \frac{2}{3})$
$(0, 1)$	$(0, 2)$
$(1, 3)$	$(1, 6)$
$(2, 9)$	$(2, 18)$

- b** State the equation of the new graph.
- c** Graph the original and new graph on the same coordinate plane.
- d** Describe the position of new graph in relation to the original graph.

40 Using a graphing calculator, graph each set of three functions below on the same screen and state whether they have the same:



i y -intercept

ii Asymptote

iii Range

a $y = e^x$, $y = e^x + 2$, and $y = e^x - 3$

b $y = e^x$, $y = e^{x-2}$, and $y = e^{x+3}$

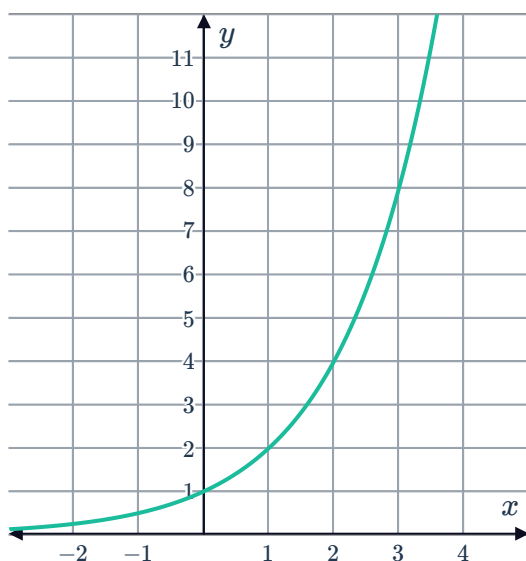
c $y = e^x$, $y = e^{2x}$, and $y = e^{3x}$

d $y = e^x$, $y = 4e^x$, and $y = -4e^x$

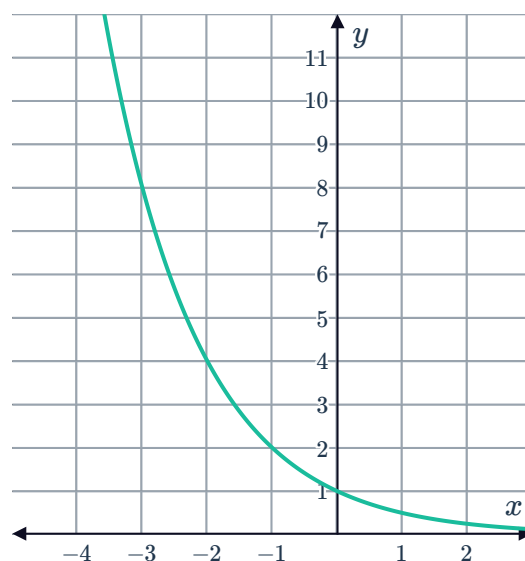
Exponential equations

41 Find the equation of the following graphs:

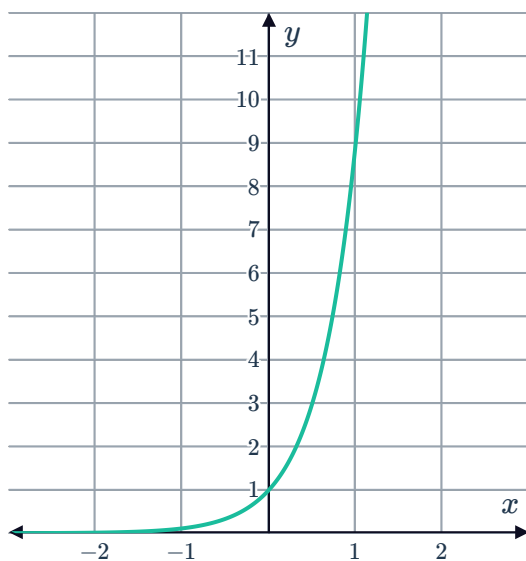
a



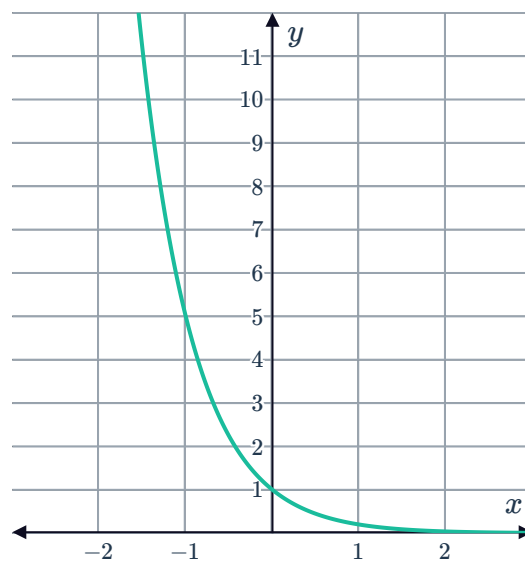
b



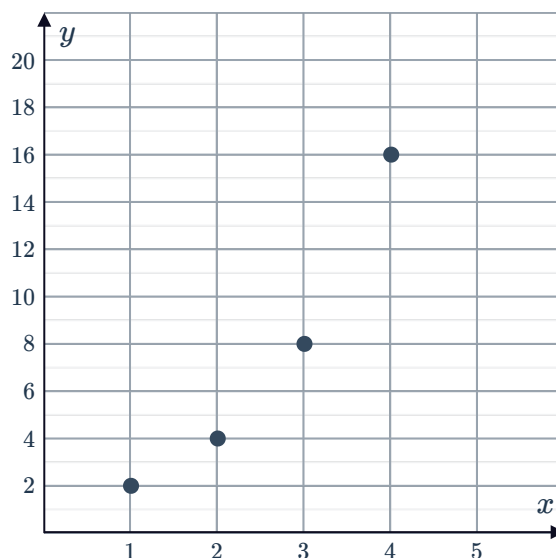
c



d



42 Consider the points given in the graph:



a Complete the table of values:

x	1	2	3	4
y				

b Identify the common ratio between consecutive y values.

c State the equation relating x and y .

43 Consider the given table of values:

x	1	2	3	4
y	10	100	1000	10 000

a Identify the common ratio between consecutive y values.

b State the equation relating x and y .

c Find the value of y when $x = 10$.

44 The points $P(-3, 320)$ and $Q(0, 5)$ satisfy an exponential function of the form $y = ab^{-x}$.

a Find the value of a .

b Find the value of b .

c Find the value of y when $x = 2$.

